MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 7049 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Reflex, pulse and CW operation, with external cavity

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er	Ik	Ehk	Pi	tk	T(seal)	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	mAdc	Vdc	W	sec	°C	ft
Maximum:	6.8	+20	350	350	-700	29	+45	12	---	250	10,000
Minimum:	5.8	-150	---	---	-15	---	-45	---	120	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Cathode:	Unipotential
Material:	Ceramic
Cooling:	Convection and conduction
Mounting position:	Any

TEST CONDITIONS:

Parameter:	Ef	Ec1	Ec2	Ec3	Er
Unit:	V	Vdc	Vdc	Vdc	Vdc
	6.3	5 to 15	300	300	Adj for max Po

GENERAL:

Qualification - Not required

1/ Replaces type ZV-1009

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1					
4229	Total reflector current	3	Er = -90 Vdc; Ec1 = 0	Ir	---	2.0	μAdc
1301	Heater current	-		If	550	650	mA
1336	Heater-cathode leakage	-	Ehk = ±45 Vdc; Ec1 = 0	Ihk	0	75	μAdc
---	Leakage	-	Ec1 = -90 Vdc; Ef = 6.3 V; no other voltages applied	Ic1/k	---	60	μAdc
1256	Electrode current (cathode)	4		Ik	---	28	mAdc
4213	Reflector voltage	4 6		Er Er	-200 -240	-330 -320	Vdc Vdc
4229	Total reflector current	4		Ir	---	50	μAdc
---	Frequency	4 6		F F	5,200 4,225	5,900 4,355	MHz MHz
1256	Electrode current (control grid)	4, 6		Ic1	0	7	mAdc
4250	Power output	4 6		Po Po	10 25	--- ---	mW mW
4214	Cathode emission	4, 5	Ef = 5.7 V	ΔIk	---	15	%
	Quality conformance inspection, part 2						
1101	Secureness of base, cap, or insert	2		---	---	---	---
1031	Low-frequency vibration	-	No voltages	---	---	---	---
4229	Total reflector current	3	Er = -90 Vdc; Ec1 = 0	Ir	---	2	μAdc
4318	Jitter	-	Ec1 = -10 Vdc; egy = 18 to 22 v; prr = 1,000; tp = 1.0 μs	t	---	0.2	μs
4280	Electronic tuning range	7		ΔF	6	---	MHz
	Quality conformance inspection, part 3						
---	Life test	4	Group C; Ec1 = 5 to 15 Vdc	t	500	---	hrs
---	Life-test end point:						
4250	Power output	4 6		Po Po	8 20	--- ---	mW mW

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Water-immersion test omitted. Torque shall be applied between upper disk (g3) and molded portion of base and shall be 20 inch-pounds.
3. Static test; tube not oscillating.
4. TUT shall be operated in the 3-3/4 mode and tested in a test cavity adjusted to resonate from 5,200 to 5,900 MHz. Adjust Er and matching network for maximum Po, adjust Ec1 for Ik = 28 mA.
5. Reduce Ef to 5.7 V while tube is oscillating. The percentage dropoff in cathode emission shall not exceed limit shown.
6. TUT shall be operated in the 2-3/4 mode and tested in a 4,290 MHz test cavity as specified on Drawing 166-JAN. Adjust Er and matching network for maximum Po.
7. Vary Er for half-power points each side of maximum, read frequency at the half-power points.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11
DSA - ES

User activities:
Army - WC, ME
Navy - AS, OS, MC, CG, SH

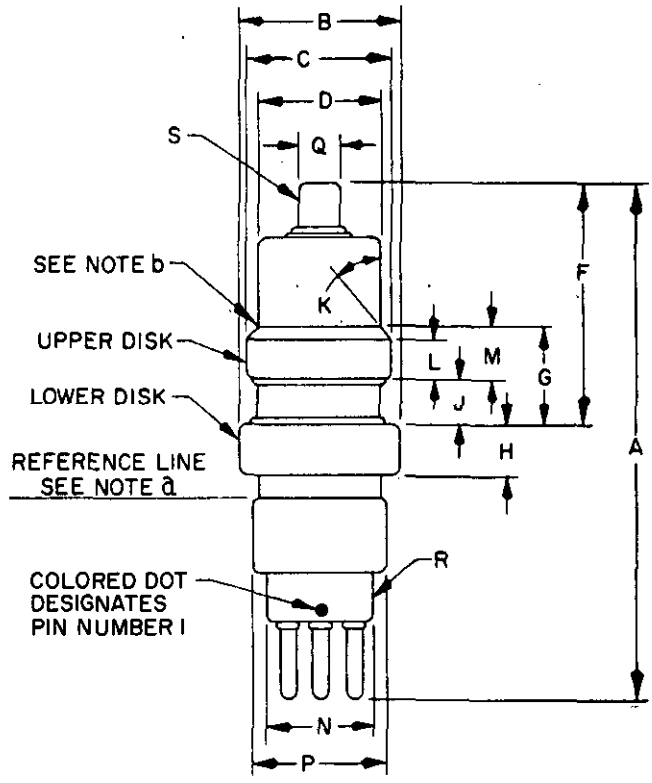
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2640)

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 1		
A	2.969 (75.41)	3.219 (81.76)
B	.990 (25.15)	1.000 (25.40)
C	.880 (22.35)	.890 (22.61)
F	1.375 (34.93)	1.500 (38.10)
M	.303 (7.70)	.323 (8.20)
Quality conformance inspection, part 2		
G	.560 (14.22)	.610 (15.49)
H	.250 (6.35)	.312 (7.92)
J	.257 (6.53)	.287 (7.29)
Quality conformance inspection, part 3 (periodic check)		
D		.750 (19.05)
K	25°	35°
N	.615(15.62)	.635(16.13)
P	.787(19.99)	.797(20.24)
Q	.245 (6.22)	.255 (6.48)
R	Base: A4-76	
S	Cap: C1-3	
Nominal dimensions		
L	.250 (6.35)	



PIN CONNECTIONS	
1	g1
2	h
3	k
4	h
Lower disk	g2
Upper disk	g3
Cap	r

NOTES:

a. For dimensions below reference line refer to base A4-76.

b. Space between ceramic and upper disk edge or backing ring shall be .030 maximum. Quality conformance inspection part 2 shall apply.

c. Diameters B and C shall be concentric within .025 TIR and shall form circular cylinder within a range of .006. Allowable radius at ends of cylinder shall be .030 maximum. Quality conformance inspection, part 2 shall apply.

d. Tube shall be tested in concentricity gage specified on Drawing 199-JAN. Gage readings on diameters shall be as follows:

Diameter	Eccentricity
D	.050 max
N	.060 max
P	.060 max
Q	.040 max

e. All concentricity measurements shall be quality conformance inspection, part 2.

f. Pin No. 1 shall be identified by an index mark (colored dot or other).

FIGURE 1. Outline drawing of electron tube type 7049.

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